

Mohammad Rezaei

M.Sc. Student in Biomedical Engineering | Computational Neuroimaging · Network Neuroscience · Biomedical AI

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EDUCATION

University of Tehran

Tehran, Iran

M.Sc. in Biomedical Engineering | GPA: **17.6/20**

Sep 2024 – Present

Thesis: *Effective Connectivity Alterations Following Motor Skill Learning: A Whole-Brain Resting-State fMRI Study.*

Group: Motor Control & Computational Neuroscience Lab.

Azad Tehran University of Medical Sciences

Tehran, Iran

B.Sc. in Biomedical Engineering | GPA: **16.24/20**

Sep 2019 – Jul 2024

Thesis: *A Comprehensive Analysis of Gray-Matter Alterations and Structural Complexity in Alzheimer's Disease: Implications for Cognitive Decline.*

PEER-REVIEWED JOURNAL PUBLICATIONS

- **Rezaei, M., Zarei, A., & Talesh Jafadideh, A.**
Neural efficiency in creative decision-making: A sparse directed subnetwork in soccer experts revealed by fMRI.
NeuroImage, 334, 121961. (2026)
DOI: [10.1016/j.neuroimage.2026.121961](https://doi.org/10.1016/j.neuroimage.2026.121961)
- **Rezaei, M., Mohammadikhavah, S., Faraji, H., Ardalani, R., Rezaei, M., & Shirazinodeh, A.**
Early diagnosis of Alzheimer's disease based on brain morphological changes: A comprehensive approach combining voxel-based morphometry and deep learning.
NeuroImage: Reports, 6(1), 100315. (2026)
DOI: [10.1016/j.ynirp.2025.100315](https://doi.org/10.1016/j.ynirp.2025.100315)

JOURNAL MANUSCRIPTS UNDER REVIEW

- **Rezaei, M., Bahrami, F., Tahmasebi Boroujeni, S., & Talesh Jafadideh, A.***
Distinct cortical–subcortical contributions underlying accuracy–speed networks following motor task learning: A resting-state fMRI study.
Scientific Reports. Status: Under review.
- **Rezaei, M. & Talesh Jafadideh, A.***
Integrated Information Decomposition reveals redundancy-related hemispheric asymmetry during negative REM dreams.
SLEEP. Status: Under review.
- **Rezaei, M., Talesh Jafadideh, A., & Bohloul, A.***
Frontal spectral entropy tracks pressure-induced decision-making failure in youth soccer.
Scientific Reports. Status: Under review.

*Corresponding author.

PEER-REVIEWED CONFERENCE PAPERS & PRESENTATIONS

- **Rezaei, M., Mohammadikhavah, S., Zarei, M., & Amiri, H.***
Interpretable skeleton-based deep learning for automated squat quality assessment in rehabilitation.
34th International Conference on Electrical Engineering (ICEE), IEEE. (May 2026)
- **Rezaei, M., Siami, M., Zarei, A., & Talesh Jafadideh, A.***

Brain network reconfiguration during creative playmaking: A task-fMRI study.

32nd National & 10th International Iranian Conf. on Biomedical Engineering (ICBME). (Nov 2025)

- **Rezaei, M.**, Talesh Jafadideh, A., Bahrami, F.^{*}, & Tahmasebi Boroujeni, S.
Effective connectivity alterations within the cortico–basal ganglia circuit associated with motor skill learning.
32nd National & 10th International Iranian Conf. on Biomedical Engineering (ICBME). (Nov 2025)

SELECTED RESEARCH PROJECTS

Human Movement Quality Assessment and Training Feedback [3D Skeleton Time Series](#) | [Rehabilitation AI](#) | [Explainable Deep Learning](#) Independent Research Project

- Built an end-to-end pipeline for analyzing human movement quality from 3D skeletal joint trajectories, including temporal alignment, body-centered normalization, bone-vector representation, and motion-derivative features.
- Compared temporal deep learning models, including CNN–BiLSTM and dilated Temporal Convolutional Networks, to classify movement quality and support automated feedback for rehabilitation and sport-training contexts.
- Integrated Grad-CAM-style temporal explanations to identify the movement phases that contributed most strongly to model decisions, improving interpretability for clinician- and coach-facing applications.

EEG Ocular Artifact Removal Using Particle Filtering [EEG Signal Processing](#) | [Bayesian Filtering](#) | [Artifact Correction](#) Independent Research Project

- Developed a Bayesian denoising framework to estimate clean EEG activity while tracking time-varying ocular contamination from EOG-related artifacts.
- Implemented particle-filtering logic for non-Gaussian and non-stationary noise conditions, where conventional linear filtering can distort neural rhythms.
- Designed the project around downstream usability, with emphasis on preserving task-relevant EEG features for time–frequency, entropy, and classification analyses.

Criticality in Excitatory–Inhibitory Neural Networks [Computational Neuroscience](#) | [Dynamical Systems](#) | [Spiking Neural Models](#) Independent Research Project

- Simulated excitatory–inhibitory spiking neural networks using Izhikevich neuron models to study synchrony, phase transitions, and avalanche-like activity patterns.
- Combined numerical simulation with mean-field and bifurcation-analysis concepts to examine how excitation–inhibition balance shapes network stability and information-processing regimes.
- Strengthened my ability to connect biological neural dynamics with computational modeling, which supports my broader interest in adaptive and human-centered training systems.

EMG Analytics for Sport and Human Performance [Surface EMG](#) | [Biomechanics](#) | [Performance Monitoring](#) Independent Research Project

- Designed an sEMG analysis workflow for studying muscle-activation patterns during sport performance, including filtering, normalization, segmentation, and feature extraction.
- Explored time-domain and frequency-domain EMG descriptors to quantify activation timing, fatigue-related changes, and coordination patterns across movement phases.
- Connected physiological signal analysis with sport-performance assessment, supporting future work on multimodal personalization using movement, muscle, and brain signals.

PROFESSIONAL EXPERIENCE

Cognitive Science & Technology Innovation Center

R&D Researcher — Affective AI

Tehran, Iran

May 2025 – Present

- Designed an affect-aware decision architecture for non-player characters using Valence–Arousal state representation, resulting in an evaluation framework now adopted for two in-house products.
- Created scenario banks spanning all four VA quadrants and multimodal congruency conditions, enabling systematic model benchmarking across 50+ culturally diverse interaction scripts.
- Authored system-specification and engineering hand-off documents used directly by NLP, speech, and computer-vision teams, reducing cross-team ambiguity and accelerating integration by one sprint cycle.

Omid Hospital

Biomedical Engineering Intern

Tehran, Iran

Dec 2022 – Jun 2023

- Performed preventive maintenance and safety checks on 15+ clinical devices (ventilators, ECG systems, infusion pumps), contributing to zero compliance violations during the inspection period.
- Surveyed robotic surgical systems and collaborated with clinical engineers on lifecycle documentation, producing a comparative report used by the hospital's procurement committee.

TECHNICAL SKILLS

Programming	Python (NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, TensorFlow, Nilearn, MNE-Python, Nipype), MATLAB, R, Bash/Linux, LaTeX, Git/GitHub, Docker.
Neuroimaging	SPM12, FSL (FEAT/MELODIC/FDT), FreeSurfer, CAT12, MRICroGL, BrainNet Viewer; task-fMRI GLM, resting-state ICA, seed-based/ROI connectivity; BIDS data management; DCM & Granger causality.
Biomedical Signals	EEG/EMG acquisition, artifact correction (ICA-based), re-referencing, time–frequency analysis, spectral entropy, information decomposition (PID/IIT), and physiological signal interpretation.
Machine Learning	CNNs, FCNs, temporal convolutional networks, recurrent models; imbalanced-data strategies, cross-validation, model calibration, ablation studies, Grad-CAM/saliency interpretability.
Statistics	GLM, LME/mixed-effects models, permutation testing, FDR/FWE correction, bootstrap confidence intervals, Cohen's d , intraclass correlation, and Bayesian model comparison.
Computer Vision	Medical image segmentation & registration, volumetric analysis, skeleton-sequence modelling, PSNR/SSIM evaluation, and clinical imaging dataset curation.

ACADEMIC SERVICE & SCIENCE COMMUNICATION

- **Peer Reviewer:** *Biomedical Signal Processing and Control* (Elsevier); *NeuroImage: Reports* (Elsevier).
- **Founder, MoScientific:** Built a free bilingual (English/Persian) educational channel covering fMRI preprocessing, GLM analysis in SPM, neuroimaging QC, and reproducible research practices.
- **Science Outreach:** Produced step-by-step tutorials and researcher interviews to lower entry barriers for neuroscience students in Iran with limited access to English-language training resources.

SELECTED COURSEWORK

Neuroscience & Biomedical	Signals & Imaging	Mathematics, Statistics & AI
Biological Signal Processing	Digital Image Processing	Machine Learning
Dynamical Systems in Neuroscience	Medical Image Analysis	Statistical Data Analysis
Biomedical Instrumentation	Signals and Systems	Linear Algebra & Engineering Mathematics
Physiology	Linear Control Systems	Stochastic Processes

CERTIFICATIONS & WORKSHOPS

- Brain Mapping, Brain Networks & Low-Intensity TMS Workshop, 2025.
- Neuroscience & Neuroimaging Specialization, *Johns Hopkins University*, Online, 2024.
- Python for Data Science, AI & Development, *IBM*, Online, 2024.
- Data Management for Clinical Research, *Vanderbilt University*, Online, 2024.
- Calibration of Medical Equipment, *Azad University, Science and Research Branch*, 2023.

LANGUAGES

Persian — Native | **English** — Professional/Academic proficiency

EXTRACURRICULAR & ATHLETICS

Competitive soccer player and team captain in local leagues; swimming and general fitness; multiple achievements at city and national levels. These experiences directly inform my research in motor control, skill acquisition, decision-making under pressure, and sports neuroscience.

REFERENCES

Prof. Fariba Bahrami Professor, Electrical & Computer Eng. University of Tehran fbahrami@ut.ac.ir	Prof. Alireza Talesh Jafadideh Professor, Engineering Science University of Tehran alireza.talesh@ut.ac.ir	Prof. Hadi Amiri Professor, Engineering Science University of Tehran hadi.amiri@ut.ac.ir
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Full contact details available upon request.